

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

The radius of a nucleus of mass number 64 is 4.8 fermi. Then the mass number of another nucleus having radius of 4 fermi is $\frac{1000}{x}$, where x is _____

Q2 - 2024 (01 Feb Shift 2)

From the statements given below :

(A) The angular momentum of an electron in n^{th} orbit is an integral multiple of h .

(B) Nuclear forces do not obey inverse square law.

(C) Nuclear forces are spin dependent.

(D) Nuclear forces are central and charge independent.

(E) Stability of nucleus is inversely proportional to the value of packing fraction.

Choose the correct answer from the options given below :

(1) (A), (B), (C), (D) only

(2) (A), (C), (D), (E) only

(3) (A), (B), (C), (E) only

(4) (B), (C), (D), (E) only

Q3 - 2024 (27 Jan Shift 1)

In a nuclear fission process, a high mass nuclide ($A \approx 236$) with binding energy 7.6 MeV/ Nucleon dissociated into middle mass nuclides ($A \approx 118$), having binding energy of 8.6 MeV/ Nucleon. The energy released in the process would be _____ MeV.

Q4 - 2024 (27 Jan Shift 2)

The atomic mass of ${}_6\text{C}^{12}$ is 12.000000u and that of ${}_6\text{C}^{13}$ is 13.003354u. The required energy to remove a neutron from ${}_6\text{C}^{13}$, if mass of neutron is 1.008665u, will be :

(1) 62.5 MeV

(2) 6.25 MeV

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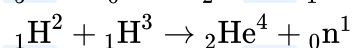
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(3) 4.95MeV

(4) 49.5MeV

Q5 - 2024 (29 Jan Shift 1)

The explosive in a Hydrogen bomb is a mixture of ${}_1\text{H}^2$, ${}_1\text{H}^3$ and ${}_3\text{Li}^6$ in some condensed form. The chain reaction is given by



During the explosion the energy released is approximately

[Given : $M(\text{Li}) = 6.01690\text{amu}$, $M({}_1\text{H}^2) = 2.01471\text{amu}$, $M({}_2\text{He}^4) = 4.00388\text{amu}$, and $1\text{amu} = 931.5\text{MeV}$]

(1) 28.12MeV

(2) 12.64MeV

(3) 16.48MeV

(4) 22.22MeV

Q6 - 2024 (30 Jan Shift 2)

In a nuclear fission reaction of an isotope of mass M , three similar daughter nuclei of same mass are formed.

The speed of a daughter nuclei in terms of mass defect ΔM will be :

$$(1) \sqrt{\frac{2c\Delta M}{M}}$$

$$(2) \frac{\Delta Mc^2}{3}$$

$$(3) c\sqrt{\frac{2\Delta M}{M}}$$

$$(4) c\sqrt{\frac{3\Delta M}{M}}$$

Q7 - 2024 (31 Jan Shift 2)

The mass number of nucleus having radius equal to half of the radius of nucleus with mass number 192 is:

(1) 24

(2) 32

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(3) 40

(4) 20

Q8 - 2024 (31 Jan Shift 2)

A nucleus has mass number A_1 and volume V_1 . Another nucleus has mass number A_2 and volume V_2 . If relation between mass number is $A_2 = 4A_1$, then $\frac{V_2}{V_1} = \underline{\hspace{2cm}}$

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Answer Key

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Q1 (27) mathongo /// math **Q2** (3) /// mathongo **Q3** (236) mathongo /// math **Q4** (3) /// mathongo

Q5 (4) mathongo /// math **Q6** (3) /// mathongo **Q7** (1) mathongo /// math **Q8** (4) /// mathongo

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Solutions

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Q1

$$R = R_0 A^{1/3}$$

$$R^3 \propto A$$

$$\left(\frac{4.8}{4}\right)^3 = \frac{64}{A}$$

$$\frac{64}{A} = 1.44 \times 1.2$$

$$A = \frac{64}{1.44 \times 1.2} = \frac{1000}{x}$$

$$x = \frac{144 \times 12}{64} = 27$$

Q2

Part of Theory

Q3

$$Q = BE_{\text{Product}} - BE_{\text{Reactant}}$$

$$= 2(118)(8.6) - 236(7.6)$$

$$= 236 \times 1 = 236 \text{ MeV}$$

Q4



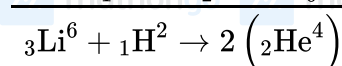
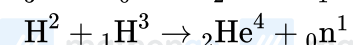
$$\Delta m = (12.000000 + 1.008665) - 13.003354$$

$$= -0.00531 \text{ u}$$

$$\therefore \text{Energy required} = 0.00531 \times 931.5 \text{ MeV}$$

$$= 4.95 \text{ MeV}$$

Q5



Energy released in process

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Solutions

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$$Q = \Delta mc^2$$

$$Q = [M(\text{Li}) + M(\text{H}^2) - 2 \times M(\text{H}_2\text{He}^4)] \times 931.5 \text{ MeV}$$

$$Q = [6.01690 + 2.01471 - 2 \times 4.00388] \times 931.5 \text{ MeV}$$

$$Q = 22.216 \text{ MeV}$$

$$Q = 22.22 \text{ MeV}$$

Q6

$$(X) \rightarrow (Y) + (Z) + (P)$$

$$M \quad M/3 \quad M/3 \quad M/3$$

$$\Delta Mc^2 = \frac{1}{2} \frac{M}{3} V^2 + \frac{1}{2} \frac{M}{3} V^2 + \frac{1}{2} \frac{M}{3} V^2$$

$$V = c \sqrt{\frac{2\Delta M}{M}}$$

Q7

$$R_1 = \frac{R_2}{2}$$

$$R_0 (A_1)^{1/3} = \frac{R_0}{2} (A_2)^{1/3}$$

$$A_1 = \frac{1}{8} A_2$$

$$A_1 = \frac{192}{8} = 24$$

Q8

For a nucleus

$$\text{Volume: } V = \frac{4}{3} \pi R^3$$

$$R = R_0 (A)^{1/3}$$

$$V = \frac{4}{3} \pi R_0^3 A$$

$$\Rightarrow \frac{V_2}{V_1} = \frac{A_2}{A_1} = 4$$

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